

Latent Structures of Urban Sustainability: A Machine Learning Analysis of Global $PM_{2.5}$ Determinants

Introduction

Urbanization has increased economic activity across the world, but it has also concentrated pollution in cities. Fine particulate matter, or $PM_{2.5}$, remains one of the most serious environmental health risks. Understanding how development relates to air quality is therefore an important policy question.

The link between income, industry, energy use, and pollution is not simple. Economic growth may support cleaner technology, yet industrial expansion can raise emissions. This report analyzes a multi-year (2008-2020) global dataset of sustainability indicators to examine these relationships. I use regression models, including neural networks, to predict $PM_{2.5}$, then apply dimensionality reduction and clustering to uncover broader patterns across cities. The goal is to understand how different development profiles are associated with urban air quality.

Data Wrangling

Data Collection

I construct a country-year panel dataset covering 2008 to 2020. The primary outcome variable is annual mean $PM_{2.5}$ concentration (EN.ATM.PM25.MC.M3), which serves as the measure of air quality throughout the analysis. All variables are drawn from the **World Bank World Development Indicators**, and I retrieved them using the World Bank API to ensure consistent formatting across countries and years (please see my code for the script). The final sample includes 154 countries with reasonably complete time series over the study period. In total, the dataset contains 13 sustainability indicators in addition to the $PM_{2.5}$ air quality measure. These include environmental measures such as **primary energy use per capita, renewable energy share, forest area, energy use per capita, and agricultural land share**; economic indicators including **GDP per capita, industry as a percent of GDP, trade as a percent of GDP, and gross capital formation**; and social and urban variables such as **urban population share, life expectancy, population density, and access to basic drinking water**. The variables were chosen based on their relevance to development patterns and environmental structure, as well as their availability from 2008 to 2020.

Merging and Cleaning

All indicators were merged using ISO3 country codes and year identifiers to construct a balanced panel dataset. Variable names were standardized to ensure consistency across datasets, and observations were ordered chronologically from 2008 to 2020 within each country. Countries with substantial amounts of missing data were excluded from the sample, resulting in the removal of 27 countries. Among the remaining countries, missing values were relatively limited, generally

accounting for less than 2 percent per variable.

Since the dataset has a time-series structure within countries, missing values were filled using **linear interpolation** within each country’s time series. This means that gaps occurring between two observed years were estimated based on the trend connecting those surrounding observations. However, missing values at the beginning or end of a country’s series were not imputed, since this would require extrapolation beyond the available data. After interpolation, any observations that still contained missing values were removed from the dataset. The final dataset contains no missing values and is fully prepared for the subsequent modeling analysis.

Urban Sustainability Index (USI)

To capture sustainability in a single measure, I construct a composite Urban Sustainability Index (USI) based on six indicators: **renewable energy share, forest area (as a share of land), access to basic drinking water, life expectancy, GDP per capita, and energy use per capita**. Each variable is rescaled to a range of 0 to 100 using min–max normalization so that differences in units do not drive the results. Since higher energy use per capita signals greater environmental pressure, its scale is reversed so that higher values reflect better sustainability performance.

The USI is calculated as the simple average of these normalized indicators. Equal weights are used to keep the index straightforward and avoid assigning subjective importance to any single component. Across the sample, the index ranges from about 31 to 75, with an average of 53.4, pointing to considerable differences in sustainability levels across countries and over time.

Note: I primarily use OECD Green Growth Indicators as my guiding point for defining the USI. For the complete data wrangling process, please refer to my code.

Exploratory Data Analysis (EDA)

Relationship Between USI and Air Quality

Figure 1 shows the country-year observations of the Urban Sustainability Index (USI) plotted against annual average $PM_{2.5}$ concentrations for the period 2008–2020. Each point corresponds to one country in a given year. A fitted linear regression line is included to illustrate the overall relationship between the two variables.

Overall, the figure suggests a negative association between sustainability and air pollution. **Countries with higher USI scores generally report lower levels of $PM_{2.5}$.** The downward trend is not especially steep, but it is visible throughout the index range. At the same time, there is considerable variation among countries with mid-range USI values. Some perform relatively well in terms of air quality, while others experience much higher pollution levels. This dispersion indicates that although sustainability is linked to cleaner air, it is not the only factor shaping pollution outcomes.

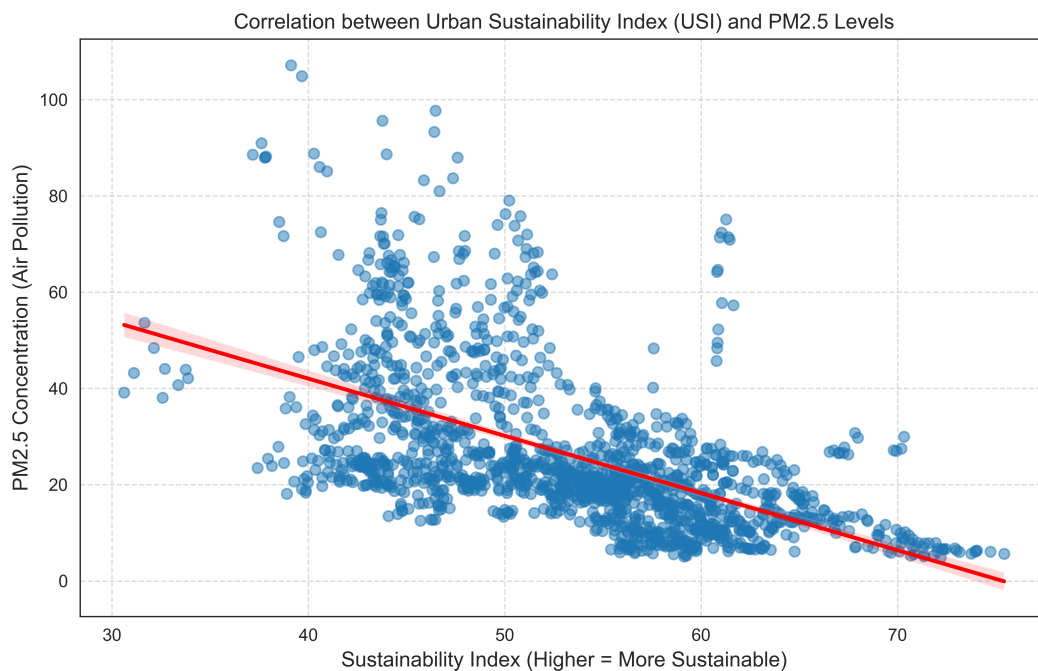


Figure 1: Scatter plot of USI versus $PM_{2.5}$ with fitted linear trend. Each point represents a country-year observation.

Global Trends Over Time

Figure 2 presents the evolution of the global average USI and average $PM_{2.5}$ concentrations between 2008 and 2020. The annual values are calculated as simple averages across all countries in the sample.

Over this period, the sustainability index shows a gradual upward trend. The increase appears fairly steady, with no obvious breaks or sudden jumps. Air pollution follows a somewhat different pattern. **Average $PM_{2.5}$ levels fluctuate in the earlier years, reach a high point around the mid-2010s, and then begin to decline toward 2020.** By the end of the sample, pollution levels are lower than they were in the middle of the period.

Taken together, the figure points to a broad improvement in sustainability alongside a modest reduction in air pollution in the later years. That said, the two trends do not move perfectly in sync, again suggesting that changes in global pollution are likely influenced by additional factors beyond sustainability alone.

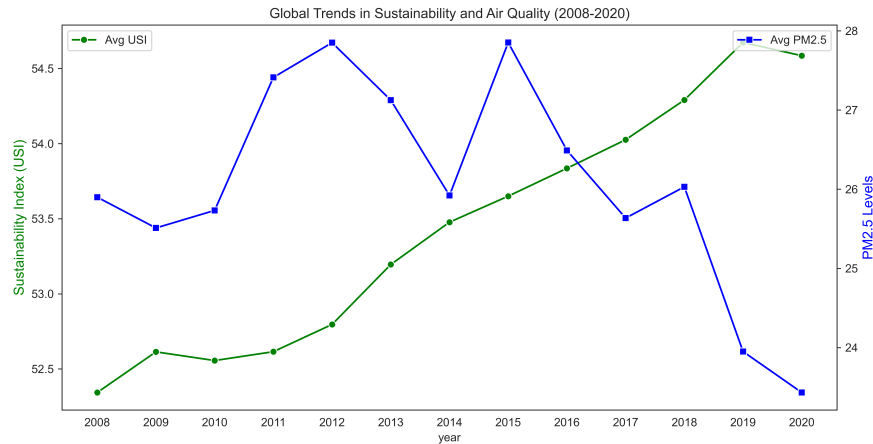


Figure 2: Global average USI and $PM_{2.5}$ levels, 2008–2020.

Cross-Country Differences in 2020

Figure 3 compares the top and bottom countries by USI in 2020. Countries are ranked by their sustainability score, and the global mean is shown for reference. The distribution reveals substantial cross-country variation. High-ranking countries cluster well above the global average, while the lowest-ranked countries remain considerably below it. The gap between the top and bottom groups exceeds 30 index points. Most top performers are high-income countries with strong institutional and environmental indicators. However, some countries (Gabon) with large natural resource endowments also rank highly, which suggests that sustainability performance does not depend solely on income level.

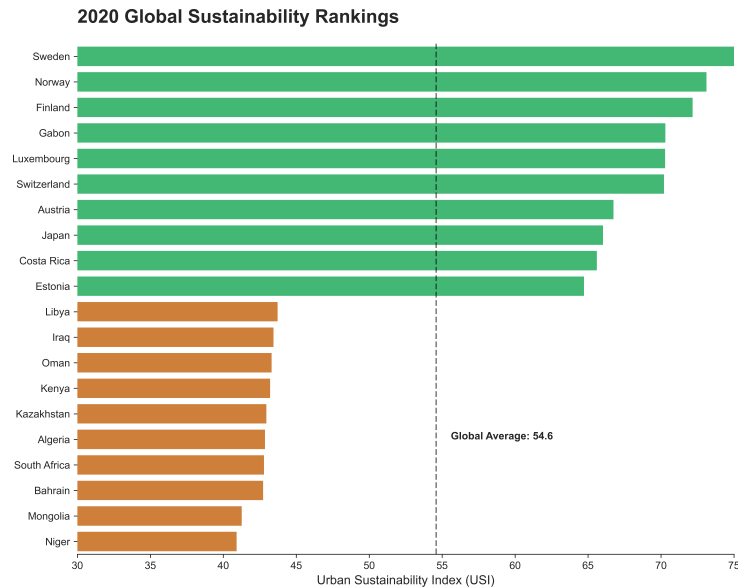


Figure 3: Top and bottom countries by USI in 2020. The dashed line indicates the global mean.

Regional Trends in $PM_{2.5}$ and USI

Figure 4 report regional averages of $PM_{2.5}$ and USI from 2008 to 2020. Regional values are computed by taking the mean across countries within each region for each year (note that due to a lack of data, South America is excluded).

Pollution levels vary substantially across regions. Europe, Oceania, and North America maintain the lowest average $PM_{2.5}$ concentrations throughout the sample, generally below $10 \mu g/m^3$. Latin America remains in the middle range, with gradual declines over time. Asia and Africa record higher pollution levels on average, though Asia shows a noticeable downward trend after the mid-2010s. The decline suggests some improvement in air quality, although levels remain elevated relative to Europe and North America.

The USI trends show gradual increases across nearly all regions over the sample period. Europe consistently records the highest sustainability scores, followed by Oceania and North America. Asia and the Middle East display steady upward movement, indicating incremental improvements. Africa shows progress as well, though at a lower overall level.

Taken together, the regional figures indicate that sustainability scores tend to rise over time in most regions, while pollution trends improve more slowly and unevenly. The regional differences observed here help explain some of the dispersion seen earlier in the USI and $PM_{2.5}$ scatter plot.

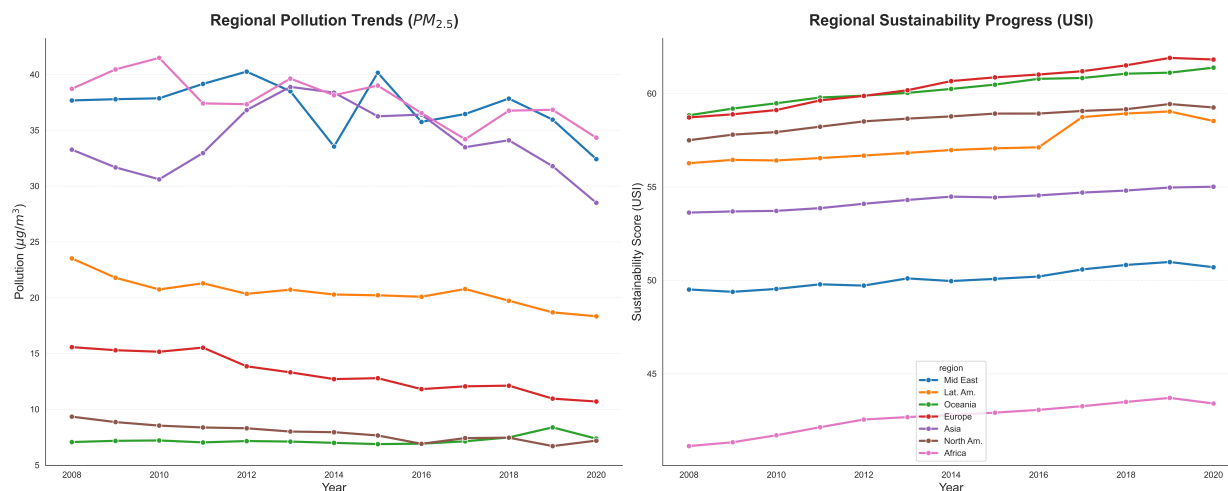


Figure 4: Regional average $PM_{2.5}$ levels and Regional average USI scores, 2008–2020.

Predictive Modeling

LASSO Regression

To predict annual $PM_{2.5}$ concentrations, I first estimate a LASSO regression model using the 13 sustainability indicators introduced earlier as predictors. **The data are split by time: observations from 2008–2017 are used for training, while 2018–2020 are held out for testing.**

This chronological split helps prevent information from future years from influencing the model estimates. All predictors are standardized using z-scores prior to estimation so that the penalty term treats them comparably.

Using only the sustainability indicators, the model achieves moderate predictive performance. The test-set R^2 is 0.379 and the mean absolute error (MAE) is 9.32. In other words, the model explains some variation in pollution levels, but a considerable portion remains unaccounted for. This suggests that sustainability indicators alone may not fully capture persistent cross-country differences in air quality.

I then extend the specification by adding regional dummy variables to account for broad geographic and climatic differences. Including these controls improves the model's performance: the test-set R^2 rises to 0.424 and the MAE falls to 8.99. The improvement implies that regional factors play a meaningful role in explaining variation in $PM_{2.5}$.

Figure 5 reports the standardized coefficients from the final LASSO model. Because all predictors are scaled, the coefficients can be compared directly in terms of magnitude. Industry share and membership in the MENA (Middle East and North Africa) region are associated with higher predicted pollution levels. In contrast, forest area, life expectancy, urban population share, and membership in the European region are linked to lower predicted $PM_{2.5}$. Several predictors have coefficients shrunk exactly to zero, showing the regularization property of LASSO, which removes variables that contribute little to predictive performance.

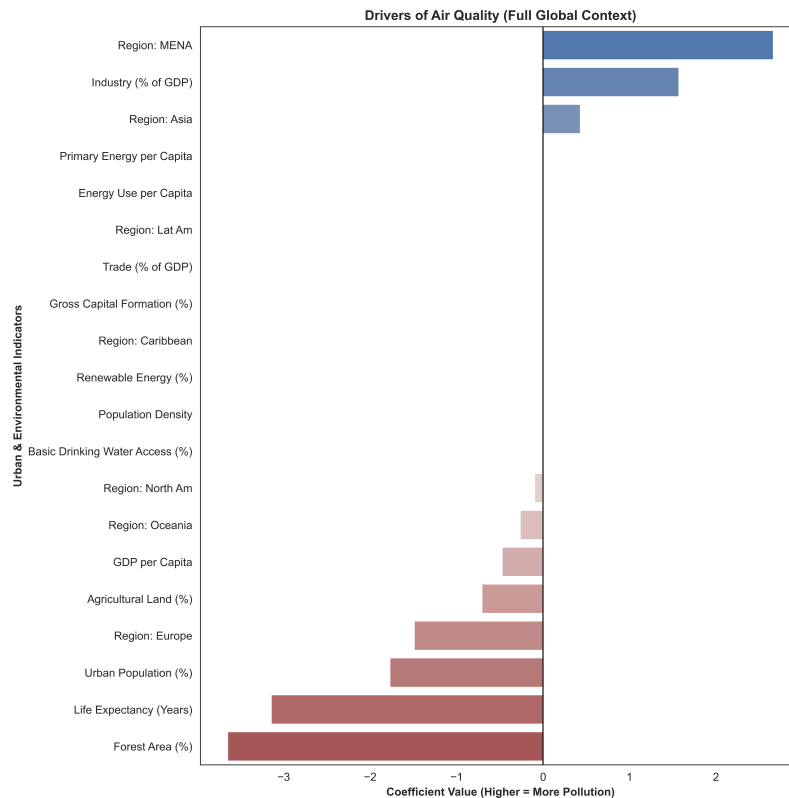


Figure 5: Standardized LASSO coefficients including regional controls. Positive values indicate higher predicted $PM_{2.5}$

Random Forest Regression

I next estimate a Random Forest model using the same chronological training–testing split as in the LASSO specification. Predictors are standardized for consistency, although tree-based methods are not sensitive to scaling. Model performance is evaluated on the held-out test set using R^2 , MAE, RMSE, median absolute error, and MAPE.

The Random Forest substantially outperforms the LASSO model. On the test set, it achieves an R^2 of 0.899 and an MAE of $3.29 \mu g/m^3$ (RMSE = 5.11; median absolute error = 2.06). The improvement indicates that the relationship between sustainability indicators and $PM_{2.5}$ is likely nonlinear and involves interactions that linear models cannot capture. Even though performance is strong, tree-based models can still overfit, so these results should be viewed as predictive rather than causal.

In Figure 6, feature importance of random forest identifies, ironically, forest area and GDP per capita as the most influential predictors, followed by population density and access to basic drinking water. Industry and energy-related variables contribute less, and regional indicators receive relatively low importance scores (see full code for regional scores; they are virtually zero).

Removing Regional Indicators

Given their low importance, I re-estimate the model using only the 13 sustainability indicators. **Test performance remains essentially unchanged ($R^2 = 0.900$, MAE = 3.28), suggesting that the sustainability variables already capture most region-related variation.** The final specification therefore excludes regional controls.

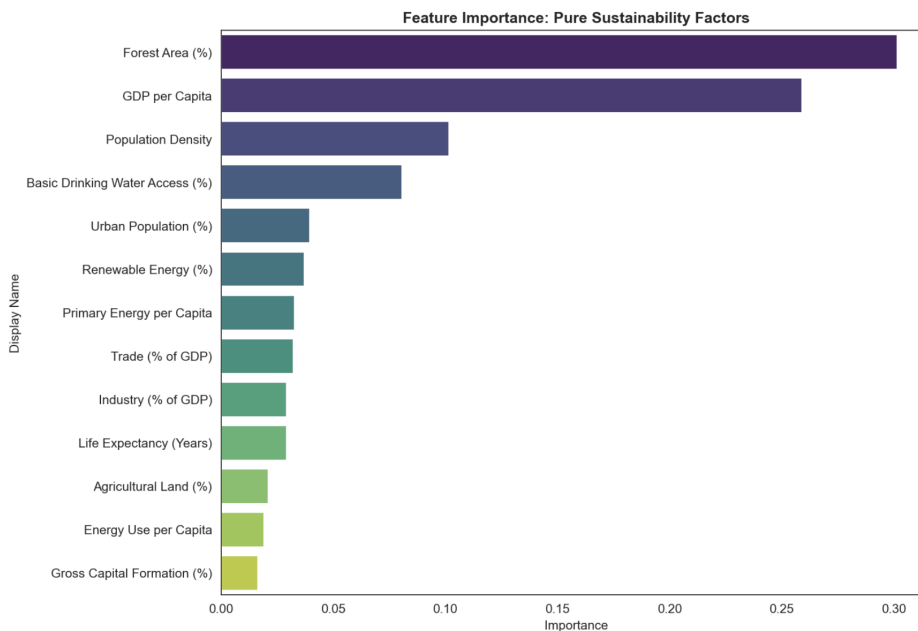


Figure 6: Random Forest feature importances for predicting $PM_{2.5}$.

Neural Network Regression (Multilayer Perceptron)

I next estimate a multilayer perceptron (MLP) to predict $PM_{2.5}$. The same chronological split is maintained, with 2008–2017 used for training and 2018–2020 reserved for testing. All predictors are standardized prior to estimation. **The baseline architecture consists of four hidden layers (128–64–32–16 units) with ReLU activation.** Early stopping is implemented to limit overfitting, and the model is trained for up to 500 iterations.

On the test set, the neural network achieves an R^2 of 0.924 and an MAE of $2.78 \mu g/m^3$ (RMSE = 4.42; median absolute error = 1.64). **Performance exceeds both LASSO and Random Forest, indicating that nonlinear interactions among sustainability indicators likely play an important role in predicting pollution levels.** As with other flexible models, however, results should be interpreted as predictive rather than causal.

Feature importance is evaluated using permutation importance on the test sample. **GDP per capita, forest area, access to basic drinking water, and life expectancy rank among the most influential predictors, with energy use and urban population share also contributing meaningfully.**

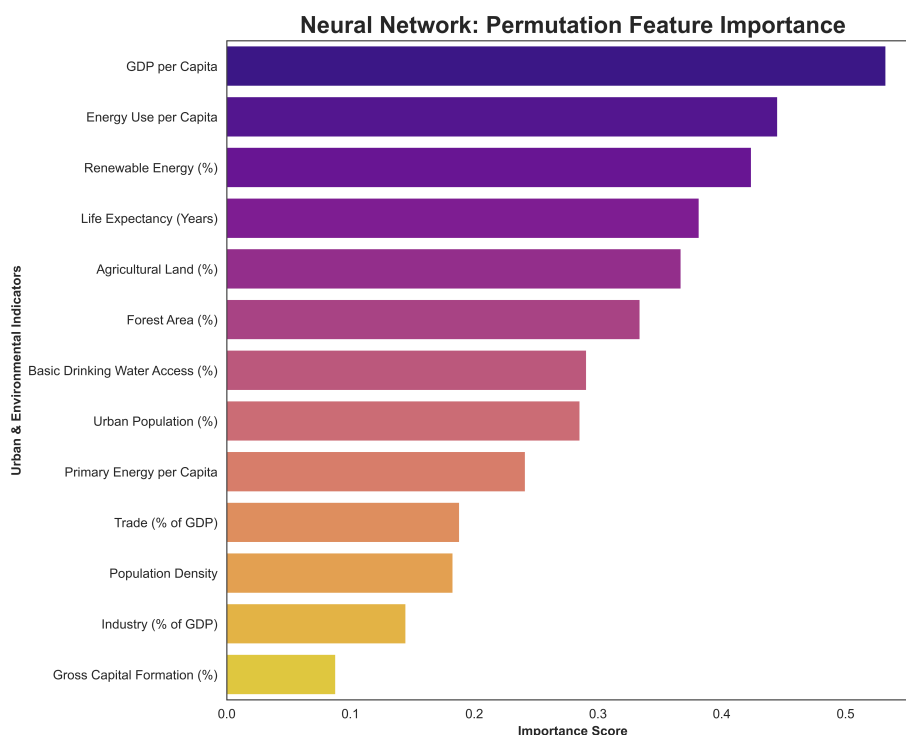


Figure 7: Permutation feature importance for the deep neural network.

Alternative Architecture: Wide Neural Network

To assess the role of model structure, I also estimate a wider but shallower MLP with a **single hidden layer of 256 units**. The focus for this run is on efficiency, as the model finishes running in 1.27 seconds. The same preprocessing, train–test split, and early stopping procedure are applied.

The model trains more quickly but reaches the maximum iteration limit before full convergence, suggesting that additional iterations or tuning might further refine performance.

This specification yields a test-set R^2 of 0.853 and an MAE of $4.13 \mu\text{g}/\text{m}^3$. While weaker than the deeper network, it still captures a substantial share of variation in $PM_{2.5}$. The comparison suggests that additional depth improves the model’s ability to approximate higher-order interactions among predictors.

Permutation importance for the wide model places relatively greater weight on life expectancy compared to GDP per capita. This shows that architectural choices influence how predictive importance is distributed across features.

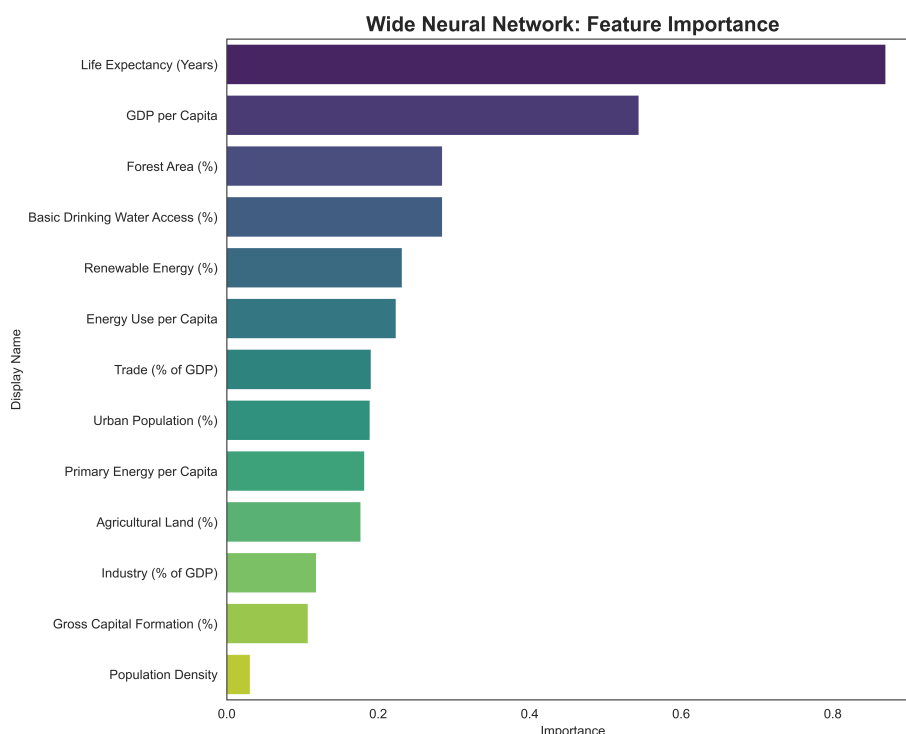


Figure 8: Permutation feature importance for the wide neural network.

Bonus Model: XGBoost Regression

As a final comparison, I estimate an XGBoost model using the same time split and evaluation framework. The model achieves a test-set R^2 of 0.884 and an MAE of $3.50 \mu\text{g}/\text{m}^3$, performing similarly to Random Forest but below the deep neural network.

Gain-based importance ranks forest area and GDP per capita as the top predictors, followed by drinking water access and population density. The fact that forest coverage and economic development consistently rank near the top across Random Forest, XGBoost, and the neural network suggests that these variables are central to predictive performance

Model Comparison

Predictive performance improves as model flexibility increases:

- LASSO: $R^2 = 0.424$
- Random Forest: $R^2 = 0.90$
- Deep MLP: $R^2 = 0.924$
- Wide MLP: $R^2 = 0.853$
- XGBoost: $R^2 = 0.884$

The pattern suggests that a purely linear specification does not fully capture the relationship between sustainability indicators and air quality. Nonlinearities and interaction effects appear to play a meaningful role. However, more flexible models carry greater overfitting risk.

Dimensionality Reduction for Regression

To explore the structure of the sustainability indicators, I apply Principal Component Analysis (PCA) after standardizing all 13 variables. The scree plot (Figure 9) shows that the first principal component explains 33.6 percent of the variance, the second 13.9 percent, and the third 10.5 percent. **Together, the first three components account for 58.0 percent of total variation.** There is no clear elbow after the first two components, and roughly six components are needed to exceed 80 percent cumulative variance. **This indicates that sustainability is not dominated by just one or two dimensions; instead, variation in the data is distributed across multiple factors.**

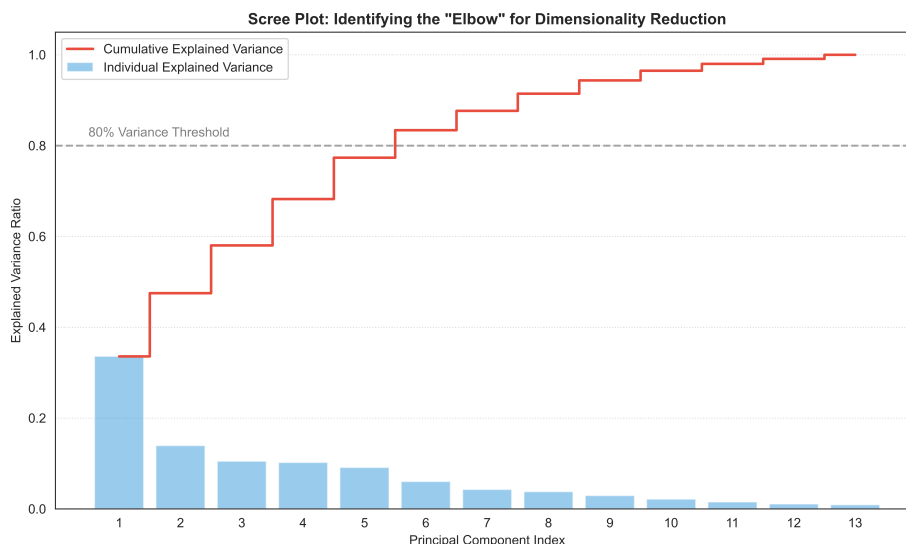


Figure 9: Scree plot of explained variance by principal component. The first three components account for 58.0% of total variance, with no clear elbow, showing variation is distributed across multiple dimensions.

Dimensionality Reduction for Unsupervised Clustering

Using the 13 sustainability indicators, I first apply PCA and then run K-Means clustering. The elbow plot (Figure 11) suggests that three clusters strike a reasonable balance: inertia declines sharply up to $K = 3$ and flattens afterwards.

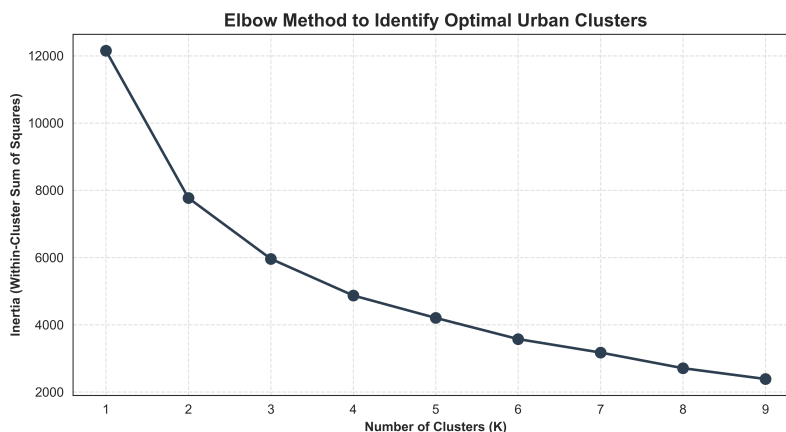


Figure 11: Elbow plot of within-cluster inertia. The flattening after $K = 3$ suggests three clusters.

Clustering is performed on the first three principal components. **PC1 captures development and modernization, loading on life expectancy, water access, GDP per capita, and urbanization.** **PC2 reflects industrial intensity, with high loadings on industry share and energy use.** **PC3 is mainly related to forest cover and land structure.**

Interpreting the Clusters

Urban Profile	PC1	PC2	PC3	Avg PM2.5
Cluster 1: Post-Industrial (Cleanest)	0.384	-0.626	-0.040	22.391
Cluster 2: Developing (Highest Pollution)	-2.748	0.636	0.184	35.666
Cluster 3: Industrialized (Moderate Pollution)	2.836	1.410	-0.135	25.216

Table 1: Urban Cluster Profiles and Principal Components

The three clusters map closely onto these dimensions:

- **Cluster 1 – Post-Industrial / Service Economies (Cleanest):** High on development, lower on industrial intensity. These countries tend to have strong infrastructure and relatively low average $PM_{2.5}$.
- **Cluster 2 – Developing / Highest Pollution:** Lower on development and moderate in industrial activity. This group shows the highest average pollution levels.

- **Cluster 3 – Industrialized / Moderate Pollution:** High on both development and industrial intensity. These countries are economically advanced but retain substantial industrial sectors, with moderate pollution levels.

The 3D PCA visualization shows clear separation along these structural dimensions (see code for interactive version). While the clusters are not perfectly isolated, they form distinct regions in the latent sustainability space.

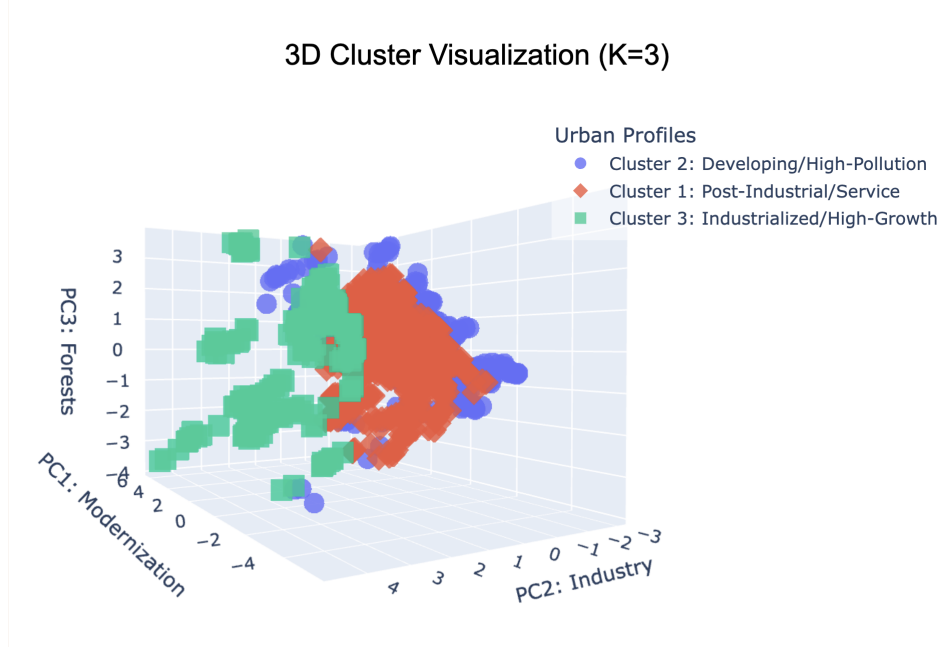


Figure 12: 3D PCA plot ($K = 3$) showing clusters separated along development, industry, and land-use dimensions.

t-SNE Validation

To test robustness, I applied t-SNE, a nonlinear dimensionality reduction method. The resulting projection shows similar group separation, even though the axes are not directly interpretable. The same three clusters emerge as distinct groupings, which supports the clustering structure from above.

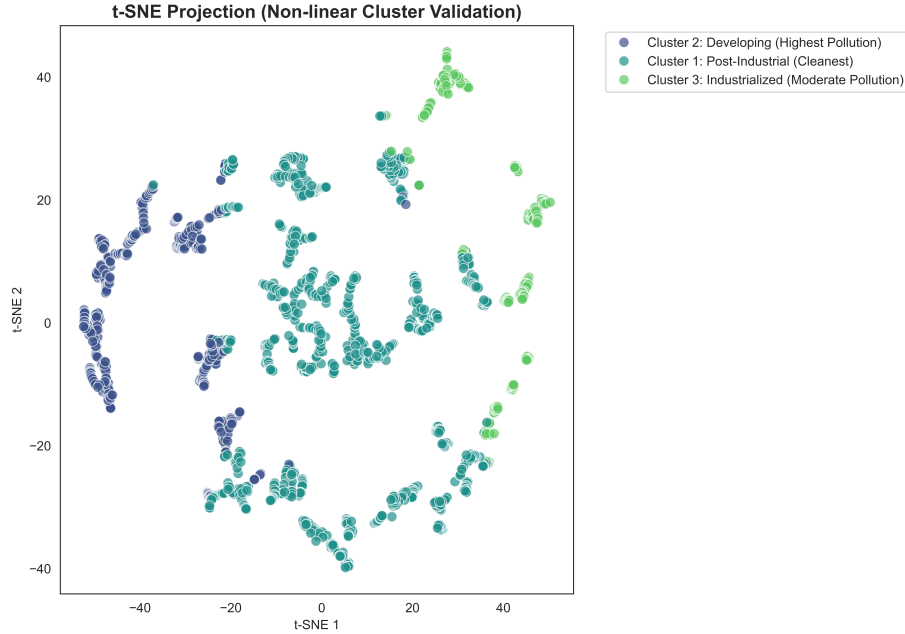


Figure 13: t-SNE projection showing similar three-cluster separation.

Cluster Analysis of Outcomes

The three clusters show clear and consistent differences in both air quality and overall sustainability performance. Refer to Figure 14.

Type 1: Clean / Modernized Service Economies

This group has the lowest average PM2.5 levels (22.39) and a mid-range Sustainability Index score (53.99). These countries tend to rely more on services than heavy industry, which likely helps keep air pollution lower.

Although they don't score the highest on sustainability overall, they appear relatively balanced. Their development model seems to support decent economic outcomes without generating excessive environmental pressure.

Type 2: Developing / Industrial Trap

This cluster stands out for having both the highest pollution levels (35.67) and the lowest Sustainability Index (49.56). These countries often combine lower levels of modernization with expanding or inefficient industrial activity.

The result is weaker sustainability capacity alongside worsening air quality. Structural challenges, such as limited infrastructure and weaker environmental regulation, may make it harder for this group to improve both economic and environmental performance at the same time.

Type 3: Large-Scale Industrialized

Type 3 countries have the highest Sustainability Index (57.34), but their pollution levels (25.22) remain moderate. These economies are highly developed and industrially strong. Their higher sustainability scores reflect higher income levels and life expectancy.

However, pollution remains higher than in Type 1. This suggests that development alone does not automatically reduce environmental pressure, especially when industrial production remains significant. They may have the potential to become more sustainable (high USI), but they are not necessarily implementing these policies.

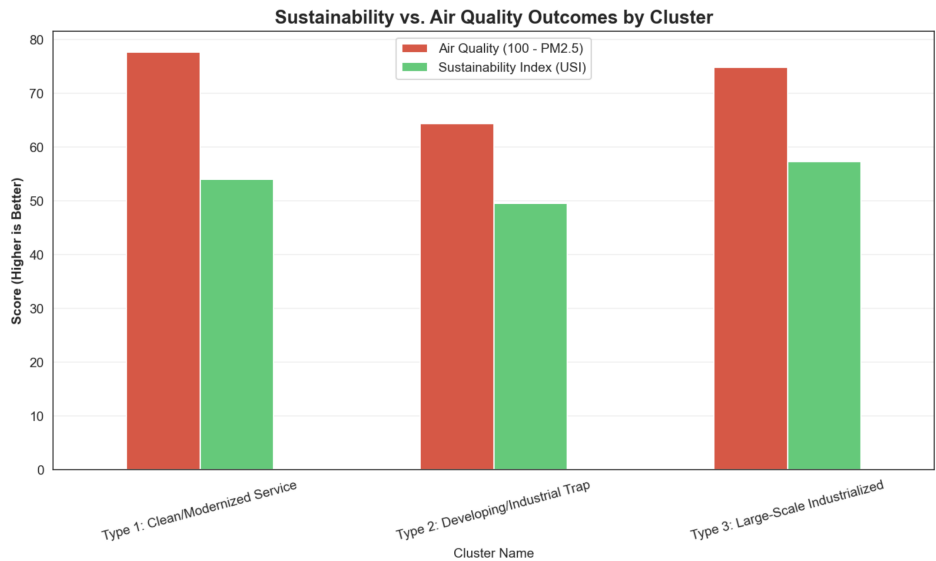


Figure 14: Average air quality (100 - PM2.5) and Sustainability Index (USI) by cluster.

Distributional Patterns

The boxplots suggest these differences aren't driven by a few outliers. For PM2.5, Type 2 shows the highest median and the widest spread, implying consistently worse and more uneven air quality. Type 1 has the lowest median and the tightest distribution, while Type 3 falls in between with moderate levels and a few outliers.

For the Sustainability Index, Type 3 has the highest median and widest range. Type 1 is slightly lower but stable, and Type 2 has the lowest median, with more countries clustered at the bottom end.

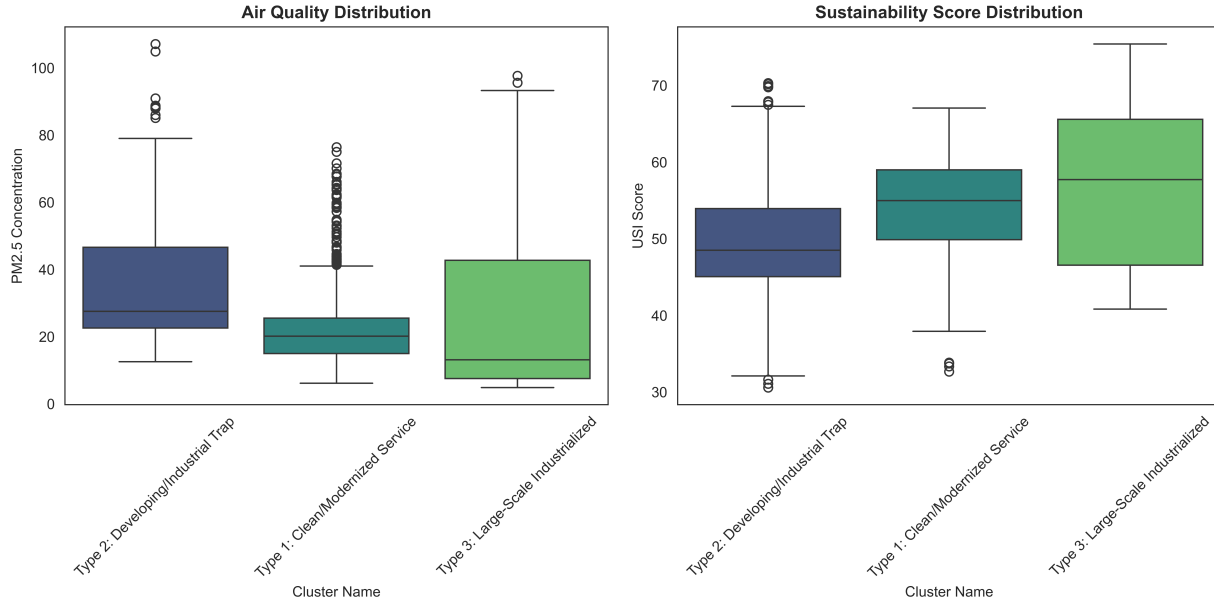


Figure 15: Distribution of PM2.5 and USI within each cluster.

Reflection on Modeling and Overfitting

Training a large number of models and then picking the one with the highest R^2 can easily turn into data snooping. If you try enough combinations, one model will eventually look impressive just by luck. In that case, it is not actually capturing a real relationship between sustainability indicators and PM2.5. It is just fitting random patterns that happen to work well for a specific split of the data. This leads to overfitting and performance numbers that are too optimistic. A model chosen this way might look strong on the sample it was trained on but fall apart when applied to new years or different countries.

To limit this risk, I used a time-based train-test split and only evaluated performance on future unseen data. I did not repeatedly adjust models based on the test results. Methods like LASSO helped keep the models simpler, and I compared tree-based models without endlessly tuning them for small gains. PCA also reduced overlap among highly correlated variables, which makes it harder for the model to chase noise. Finally, I checked whether the main findings were consistent across different modeling approaches. Since similar predictors and cluster patterns kept showing up, it gives more confidence that the results reflect real structure rather than overfitting.

Conclusion

In this project, I examined how development, economic structure, and environmental factors relate to air pollution across countries. Using 13 sustainability indicators, I first built predictive models, and then I reduced dimensionality with PCA while applying clustering to identify broader country profiles. **Across all parts of the analysis, development level and industrial structure consistently showed up as key drivers of PM2.5.**

Nonlinear models performed much better than linear ones, suggesting the relationship between sustainability indicators and pollution is complex. Variables like forest area, GDP per capita, life expectancy, and industry share repeatedly appeared as important predictors. When I reduced the data using PCA, predictive performance dropped, which shows that pollution depends on several interacting factors rather than just one or two broad dimensions.

The clustering analysis reinforced this pattern. **Countries are grouped into three types: cleaner service-based economies, developing countries with high pollution, and industrialized but economically strong countries with moderate pollution.** This highlights how long-term improvements in air quality depend not just on environmental policy, but also on deeper structural changes in the economy.